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IMPLANTABLE MEDICAL DEVICE WITH PACING CAPTURE CLASSIFICATION

Abstract

This disclosure is directed to devices and techniques for classifying of pacing captures to evaluate effectiveness of pacing by a pacing device, such as an implantable medical device (IMD). An example system includes stimulation circuitry to generate a pacing stimulus, sensing circuitry to sense an evoked response after the pacing stimulus, and processing circuitry. The processing circuitry determines classification features from the evoked response and applies the classification features to a classification model, the classification model generated by a machine learning algorithm using one or more test sets comprising a plurality of sample evoked responses for each of a plurality of classifications. Based on the output of the model, the processing circuitry classifies the evoked response as one of the plurality of classifications.

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Background/Summary

RELATED APPLICATION [0001] This application is a continuation of U.S. application Ser. No. 17/401,185 filed Aug. 12, 2021, which claims the benefit of U.S. Provisional Application No. 63/072,685, filed Aug. 31, 2020, the entire contents of each of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] This disclosure generally relates to medical devices, and more specifically, delivery of cardiac pacing by medical devices.

BACKGROUND

[0003] Cardiac pacing is delivered to patients to treat a wide variety of cardiac dysfunctions. Cardiac pacing is often delivered by an implantable medical device (IMD). An implantable cardioverter-defibrillator (ICD), for example, may provide pacing functionality and also provide cardioversion or defibrillation in response to detected cardiac tachyarrhythmias, if needed. An IMD typically delivers such therapy to the heart via electrodes located on one or more leads, which may be intracardiac or extracardiovascular leads, although leadless IMDs for delivering such therapies have also been implemented. Patients with heart failure may be treated with cardiac resynchronization therapy (CRT). CRT is a form of cardiac pacing. The ventricles of some heart failure patients contract in an uncoordinated, or asynchronous, manner, which greatly reduces the pumping efficiency of the ventricles. CRT delivers pacing pulses at particular times, e.g., atrioventricular (A-V) intervals and/or intra-ventricular (V-V) intervals, and particular locations, e.g., to one or both of the right and left ventricles, to re-coordinate the contraction of the ventricles. In some examples, CRT involves delivery of pacing pulses to both ventricles to synchronize their contraction. In other examples, CRT involves delivery of pacing pulses to one ventricle, such as the left ventricle, to synchronize its contraction with that of the right.

SUMMARY

[0004] In general, the disclosure is directed to devices and techniques for classifying of pacing captures to evaluate effectiveness of pacing by a pacing device, such as an implantable medical device (IMD). The techniques described here use the physiologically meaningful features of an evoked response to pacing to classify the capture of cardiac tissue by the pacing delivered by the pacing device. These features can be measured or derived from the sample evoked response.

[0005] Machine learning is used to produce a model that can be run on the relatively limited processing capacity of an IMD. Evoked responses may be categorized to reflect the effectiveness of pacing. For example, evoked responses may be categorized, in order of effectiveness, as “selective capture,” “non-selective capture,” “right ventricular (RV) capture,” or “no ventricular capture.” Samples of evoked responses are collected for each classification. In some examples, these sample evoked responses may include different measurements made by an implantable devices, such as, far field (FF) measurements, near field measurements (NF), and differential (DIFF) measurements. To prepare for model generation, the physiologically meaningful features are calculated and/or derived for each sample. These prepared samples are used to generate one or more training sets and one or more test sets. The training sets used to train a machine learning algorithm to classify pacing captures. Different machine algorithms may be used. For example, a classification algorithm or a

regression algorithm may be used. In some examples, a classification machine learning algorithm may produce a decision tree to classify the pacing based on the features of the evoked response. The decision tree may, in some examples, identify key features from the physiologically meaningful features that are used to classify the pacing capture.

[0006] The model is downloaded to the IMD with the physiologically meaningful features that are used to classify the pacing. Periodically (e.g., hourly, daily, weekly, monthly, etc.), the IMD implements a pacing test to evaluate the current pacing settings and, when necessary, adjust the pacing settings. The pacing test provides pacing to the heart and the IMD captures the resulting evoked responses. IMD extracts the physiologically meaningful features from the evoked responses and implements the model to classify the pacing. Based on the classification, the implantable device may adjust, for example, the voltage level of the stimulation provided by an electrode of the IMD. For example, patient experience and/or device performance may be better when a relatively low voltage is used by the IMD while still providing “selective capture” pacing. In some examples, the voltage provided in the pacing stimulation may be adjusted during the pacing test to determine whether a lower voltage will still provide “selective capture” pacing. In this example, when a lower stimulation voltage can be used to provide effective therapy, the patient receives the therapy while the battery life of the IMD increases.

[0007] An example system includes stimulation circuitry configured to generate a pacing stimulus, sensing circuitry configured to sense an evoked response after the pacing stimulus, and processing circuitry. The processing circuitry is configured to determine classification features from the evoked response and apply the classification features to a classification model, the classification model generated by a machine learning algorithm using one or more test sets comprising a plurality of sample evoked responses for each of a plurality of classifications. Based on the output of the model, the processing circuitry is configured to classify the evoked response as one of the plurality of classifications.

[0008] An example method comprises generating a pacing stimulus and sensing an evoked response after the pacing stimulus. The method also includes determining classification features from the evoked response. The method also includes applying the classification features to a classification model. The classification model is generated by a machine learning algorithm using one or more test sets comprising a plurality of sample evoked responses for each of a plurality of classifications. Additionally, the method includes, based on the output of the model, classifying, by the processing circuitry, the evoked response as one of the plurality of classifications.

[0009] An example computer readable medium comprising instructions, that when executed, cause an implantable medical device (IMD) to generate, by stimulation circuitry, a pacing stimulus, and sense, by sensing circuitry, an evoked response after the pacing stimulus. The instructions also cause the IMD to determine, by processing circuitry, classification features from the evoked response, and apply the classification features to a classification model. The classification model is generated by a machine learning algorithm using one or more test sets comprising a plurality of sample evoked responses for each of a plurality of classifications. Further, the instructions cause the IMD to, based on the output of the model, classify, by the processing circuitry, the evoked response as one of the plurality of classifications.

[0010] The summary is intended to provide an overview of the subject matter described in this disclosure. It is not intended to provide an exclusive or exhaustive explanation of the systems, device, and methods described in detail within the accompanying drawings and description below. Further details of one or more examples of this disclosure are set forth in the accompanying drawings and in the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] FIG. 1 is a conceptual diagram illustrating an example system for classifying evoked responses to determine an effectiveness of pacing, in accordance with the teachings of this disclosure.

[0012] FIG. 2 is a conceptual diagram illustrating the IMD and leads of the system of FIG. 1 in greater detail.

[0013] FIG. 3 is a block diagram illustrating an example configuration of an IMD for classifying evoked responses to determine an effectiveness of pacing, in accordance with the teachings of this disclosure.

[0014] FIG. 4 depicts an example evoked response illustrating physiologically meaningful features that may be extracted from the evoked response and used to classify the evoked response, in accordance with the teachings of this disclosure.

[0015] FIGS. 5A, 5B, 5C, and 5D depict example classifications for evoked responses that were classified in accordance with the teachings of this disclosure.

[0016] FIG. 6 is a conceptual diagram illustrating an example system to generate the classification model of FIG. 1 using supervised machine learning, in accordance with the teachings of this disclosure.

[0017] FIG. 7 is a flowchart of an example method to classify evoked responses, in accordance with the teachings of this disclosure.

[0018] FIG. 8 is a is a conceptual diagram of a leadless intracardiac pacemaker positioned within the right atrium for providing ventricular pacing via the His bundle, in accordance with the teachings of this disclosure.

DETAILED DESCRIPTION

[0019] This disclosure is directed to devices and techniques for classifying of pacing captures to evaluate effectiveness of pacing by a pacing device, such as an implantable medical device (IMD). As used herein, “pacing” refers to the delivery of an electrical impulse and “capture” refers to depolarization of myocardial cells by an electrical impulse. An IMD, such as a pacemaker, is connected to electrodes that are in contact with heart muscle. Pacing by the implantable device at the electrodes cause capture to occur in a wave across the heart muscle. Generally, this depolarization wave causes the heart to beat. From time to time, shortly after delivering the electrical stimulation (e.g., within 90 to 120 milliseconds, etc.), the IMD measures an evoked electrical response of the heart muscle. The morphology of the evoked response is indicative of which parts of the heart were affected by the pacing and in which sequence they were affected. This indicates the effectiveness of the pacing (sometimes referred to as “beat truthing”).

[0020] The IMD may classify the response of the heart muscle to the pacing based on the characteristics of the evoked response to determine the effectiveness of the pacing. Pacing, as evidenced by the evoked response, can be classified into several categories. For example, for His bundle pacing (HBP), or other conduction system pacing, the pacing may be classified as “selective capture,” “non-selective capture,” “right ventricular (RV) capture,” or “no ventricular capture.” Generally, selective capture is preferable to non-selective capture (which is preferable to RV capture), while no ventricular capture is undesirable. Selective capture, for example, implies capture of the His bundle alone with resulting conduction via the His-Purkinje axis, leading to a QRS duration and morphology identical (or substantially identical) to the patient's native QRS duration. As another example, non-selective capture implies the additional capture of the septal myocardium, resulting in right ventricular (RV) myocardial pre-excitation and initial slurring and widening of the QRS complex. RV pacing implies excitation primarily of the RV myocardium and a widening of the QRS complex.

[0021] As described below, the IMD uses physiologically meaningful features of the evoked response to pacing to classify the pacing. These features can be measured or derived from the

evoked responses. Machine learning is used to produce a model that can be run on the relatively limited processing capacity of an IMD. Additionally or alternatively, the model may run off-line or in a cloud. In some examples, the model may be optimized (e.g., the training set customized) based on a certain population (e.g., a shared location and/or a shared set of demographic and physical data, etc.) and/or individual patient data. The classification of the evoked responses may reflect the effectiveness of pacing. Samples of evoked responses are collected for each classification. In some examples, these sample evoked responses may include different measurements made by an implantable device, such as, far field (FF) measurements, near field measurements (NF), and differential (DFF) measurements. To prepared for model generation, the physiologically meaningful features are calculated and/or derived for each sample. These prepared samples are used to generate one or more training sets and one or more test sets. The training sets used to train a machine learning algorithm to classify pacing captures. Different machine algorithms may be used, such as a classification algorithm or a regression algorithm, or an ensemble learning method for classification and regression (e.g., random forests or random decision forests). In some examples, a classification machine learning algorithm may produce a decision tree model to classify the pacing based on the features of the evoked response. The decision tree may, in some examples, identify key features from the physiologically meaningful features that are used to classify the pacing.

[0022] The model is downloaded to the IMD with the physiologically meaningful features that are used to classify the pacing. Periodically (e.g., daily, weekly, monthly, etc.), the IMD implements a pacing test to evaluate the current pacing settings and, when necessary, adjust the pacing settings. The pacing test provides pacing to the heart and IMD captures the resulting evoked responses. IMD extracts the physiologically meaningful features from the evoked responses and implements the model to classify the pacing. Based on the classification, the implantable device may adjust, for example, the voltage level of the stimulation provided by an electrode of the IMD. For example, patient experience may be better when a relatively low voltage is used by the IMD while still providing “selective capture” pacing. In some examples, the voltage provided in the pacing stimulation may be adjusted during the pacing test to determine whether a lower voltage will still provide “selective capture” pacing. In this example, when a lower stimulation voltage can be used to provide effective therapy, the patient receives the therapy while the battery life of the IMD increases.

[0023] FIG. 1 is a conceptual diagram illustrating an example system 10 for classifying evoked responses to determine an effectiveness of pacing. As illustrated by example system 10 in FIG. 1, a system for classifying evoked responses according to the techniques of this disclosure may include an implantable medical device (IMD) 16 with pacing capabilities. IMD 16 is connected to lead 22 and is communicatively coupled to external device 24. IMD 16 senses electrical signals attendant to the depolarization and repolarization of heart 12, e.g., an EGM, via electrodes on one or more of lead 22 and/or the housing of IMD 16. IMD 16 may also deliver therapy in the form of electrical signals to heart 12 via electrodes located on lead 22. The therapy may be pacing, cardioversion and/or defibrillation pulses. IMD 16 may monitor EGM signals collected by electrodes on lead 22, and based on the EGM signal, classify the pacing to determine its quality.

[0024] Lead 22 (sometime referred to as a “His bundle lead”) extends into heart 12 of patient 14 to sense electrical activity of heart 12 and/or deliver electrical stimulation to heart 12. In the example shown in FIG. 1, lead 22 extends through one or more veins (not shown), the superior vena cava (not shown), and right atrium 26, and into right ventricle 28. IMD 16 may include additional leads, such as a left ventricular (LV) that extends through one or more veins, the vena cava, right atrium 26, and into the coronary sinus to a region adjacent to the free wall of left ventricle 32 of heart 12. Lead 22 is positioned to provide pacing to the His bundle 20 (providing pacing at this location is sometimes referred to as “His bundle pacing” or “HBP”). In the illustrated example, lead 22 is positioned to provide pacing to the His bundle 20 between an atrioventricular node (not shown) and branches of Purkinje fibers 30.

[0025] Quality and, in some examples, effectiveness of pacing may be determined by the reaction of heart **12** to pacing. His bundle **20** efficiently transmits impulses from the atrioventricular node of heart **12** to the ventricles (e.g., right ventricle **28** and left ventricle **32**) of heart **12** via Purkinje fibers **30**. The depolarization of myocardial cells caused by HBP may resemble the depolarization of myocardial cells caused by natural pacing. The capture resulting from pacing that primarily affects His bundle **20** is referred to as “selective capture.” In some examples, pacing may cause additional capture of the septal myocardium that results in right ventricular (RV) myocardial pre-excitation (e.g., stimulus affects His bundle **20** and additional septal myocardium, etc.). The resulting capture may be referred to as “non-selective capture.” Capture that is primarily of the septal myocardium of the right ventricle may be referred to as “RV capture.” Pacing that does not affect the His bundle or the septal myocardium of the right ventricle may be said to have caused “no ventricular capture.”

[0026] In some examples, external device **24** takes the form of a handheld computing device, computer workstation or networked computing device that includes a user interface for presenting information to and receiving input from a user. A user, such as a physician, technician, surgeon, electro-physiologist, or other clinician, may interact with external device **24** to retrieve physiological or diagnostic information from IMD **16**. A user may also interact with external device **24** to program IMD **16**, e.g., select values for operational parameters of the IMD. External device **24** may include processing circuitry configured to evaluate EGM signals transmitted from IMD **16** to external device **24**.

[0027] IMD **16** and external device **24** may communicate via wireless communication using any techniques known in the art. Examples of communication techniques may include, for example, low frequency or radiofrequency (RF) telemetry, or according to the Bluetooth® or Bluetooth LE specifications. In some examples, external device **24** may be located remotely from IMD **16** and communicate with IMD **16** via a network.

[0028] System **10** of FIG. **1** is an example of a system for classifying evoked responses resulting from pacing to determine an effectiveness and/or quality of pacing. Processing circuitry of IMD **16** includes pacing analysis circuitry **34** a classification model **35** configured to classifying evoked responses resulting from pacing. From time to time, shortly after delivering the electrical stimulation (e.g., within 90 to 120 milliseconds, etc.), the cardiac signal analysis circuitry measures an evoked electrical response (sometimes referred to as an “evoked response”) characterizing the depolarization wave of the heart muscle of heart **12**. The evoked response is measured via a cardiac electromyogram (EGM) sensed via one or more electrodes of IMD **16**. The cardiac signal analysis circuitry may measure far field (FF) potential (e.g., from RVtip electrode **25A** of lead **22** to the exterior of IMD **16**) and near field (NF) potentials (e.g., RVtip electrode **25A** to RVring electrode **25B** of lead **22**). The cardiac signal analysis circuitry may also determine differential far field (DFF) potential (e.g., a first order differential of the FF potential, etc.). For example, a cardiac EGM that includes an evoked response may include P-waves (depolarization of the atria), R-waves (depolarization of the ventricles), and T-waves (repolarization of the ventricles), among other events. As described below, the cardiac EGM of an evoked response may include physiologically meaningful features. These physiologically meaningful features are extracted by pacing analysis circuitry **34**. Pacing analysis circuitry **34** then uses classification model **35** to classify the evoked response. In some examples, IMD **16** may take actions based on the classification of the evoked response, such as adjusting pacing parameters and/or providing an alert (e.g., to external device **24**).

[0029] Although the techniques for classifying evoked responses resulting from pacing to determine an effectiveness and/or quality of pacing according to the techniques of this disclosure are described herein primarily with reference to example system **10**, the techniques may be performed by other systems that differ from example system **10**. For example, systems for identifying the one or more parameters according to the techniques of this disclosure may include

an IMD having different functionality than IMD **16**, and may include more, fewer, or different implantable cardiac leads than lead **22**. In some examples, systems for identifying the one or more parameters include more or fewer leads, do not include any intracardiac leads, or do not include any leads. Example IMDs that may implement the techniques of this disclosure in addition to the illustrated example of IMD **16** include intravascular or extracardiovascular ICDs, and transcatheter pacing systems, such as the Micra™ transcatheter pacing system commercially available from Medtronic plc, of Dublin, Ireland.

[0030] FIG. **2** is a conceptual diagram illustrating IMD **16** and lead **22** of system **10** in greater detail. In the illustrated example, bipolar electrodes **25A** is located adjacent to a distal end of lead **22**. In the illustrated example, electrode **25A** may take the form of extendable helix or tine tip electrodes. Electrode **25B** may take the form of a ring electrode electrically insulated from tip electrode **25A**. In some examples, each of electrodes **25A** and **25B** is electrically coupled to a respective conductor within the lead body of lead **22** and thereby coupled to circuitry within IMD **16**.

[0031] In some examples, IMD **16** includes one or more housing electrodes, such as housing electrode **4** illustrated in FIG. **2**, which may be formed integrally with an outer surface of hermetically sealed housing **8** of IMD **16** or otherwise coupled to housing **8**. In some examples, housing electrode **4** is defined by an uninsulated portion of an outward facing portion of housing **8** of IMD **16**. Other divisions between insulated and uninsulated portions of housing **8** may be employed to define two or more housing electrodes. In some examples, a housing electrode includes substantially all of housing **8**.

[0032] Housing **8** encloses signal generation circuitry that generates therapeutic stimulation, such as cardiac pacing, cardioversion, and defibrillation pulses, as well as sensing circuitry for sensing electrical signals attendant to the depolarization and repolarization of heart **12**. As described below, housing **8** may also enclose processing circuitry and memory configured with pacing analysis circuitry **34** and classification model **35**. Housing **8** may also enclose telemetry circuitry for communication between IMD **16** and external device **24**.

[0033] IMD **16** senses electrical signals attendant to the depolarization and repolarization of heart **12** via electrodes **4**, **25A**, and **25B**. Any of the electrodes **25A** and **25B** may be used for unipolar sensing in combination with housing electrode **4**. The illustrated numbers and configurations of lead **22** and electrodes **25A** and **25B** are merely examples. Other configurations, i.e., number and position of leads and electrodes, are possible. In some examples, system **10** may include an additional lead or lead segment having one or more electrodes positioned at different locations in the cardiovascular system for sensing and/or delivering therapy to patient **14**. For example, instead of or in addition to intracardiac lead **22**, system **10** may include one or more extracardiovascular (e.g., epicardial, substernal, or subcutaneous) leads not positioned within the heart.

[0034] FIG. **3** is a block diagram illustrating an example configuration of IMD **16**. In the illustrated example, IMD **16** includes processing circuitry **70**, memory **72**, signal generation circuitry **74**, sensing circuitry **76**, telemetry circuitry **78**, pacing analysis circuitry **34**, and activity sensor **82**. Memory **72** includes computer-readable instructions that, when executed by processing circuitry **70**, cause IMD **16** and processing circuitry **70** to perform various functions attributed to IMD **16** and processing circuitry **70** herein. Memory **72** may include any volatile, non-volatile, magnetic, optical, or electrical media, such as a random access memory (RAM), read-only memory (ROM), non-volatile RAM (NVRAM), electrically-erasable programmable ROM (EEPROM), flash memory, or any other digital or analog media.

[0035] Processing circuitry **70** may include any one or more of a microprocessor, a controller, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or equivalent discrete or analog logic circuitry. In some examples, processing circuitry **70** may include multiple components, such as any combination of one or more microprocessors, one or more controllers, one or more DSPs, one or more ASICs, or

one or more FPGAs, as well as other discrete or integrated logic circuitry. The functions attributed to processing circuitry **70** herein may be embodied as software, firmware, hardware, or any combination thereof.

[0036] Signal generation circuitry **74** is configured to generate and deliver pacing stimulation to patient **14**. As shown in FIG. **3**, signal generation circuitry **74** is electrically coupled to electrodes **4**, **25A**, and **25B**, e.g., via conductors of lead **22** and, in the case of housing electrode **4**, within housing **8**. For example, signal generation circuitry **74** may deliver pacing, defibrillation, or cardioversion pulses to heart **12** via at least electrode **25A**. In some examples, signal generation circuitry **74** delivers therapy in the form of signals other than pulses such as sine waves, square waves, or other substantially continuous time signals.

[0037] Electrical sensing circuitry **76** monitors electrical cardiac signals from any combination of electrodes **4**, **25A**, and **25B**. Sensing circuitry **76** may also include switching circuitry which processing circuitry **70** controls to select which of the available electrodes are used to sense the heart activity, depending upon which electrode combination is used in the current sensing configuration. In some examples, sensing circuitry **76** may include one or more amplifiers, filters, and analog-to-digital converters.

[0038] Sensing circuitry **76** may include one or more detection channels, each of which may include an amplifier. The detection channels may be used to sense cardiac signals, such as cardiac EGMs indicative of an evoked response after signal generation circuitry **74** generates the pacing stimulation. For example, sensing circuitry **76** may sense an EGM of the NF potential and an EGM of the FF potential. Sensing circuitry **76** may apply a low-pass filter (e.g., 12 Hertz (Hz) at a 256 Hz sampling rate, etc.) and/or a five-point differential (e.g., at a 256 Hz sampling rate, etc.). The detection channels may provide the signals to an analog-to-digital converter, for conversion into a digital signal for processing or analysis by processing circuitry **70** or pacing analysis circuitry **34**.

[0039] Periodically (e.g., daily, weekly, monthly, etc.), processing circuitry **70** may perform a series of test pacing to evaluate the pacing being provided to heart **12**. During these series of test pacing, pacing analysis circuitry **34** causes sensing circuitry **76** to capture cardiac EGMs after signal generation circuitry **74** generates the pacing stimulation for the pacing tests. Pacing analysis circuitry **34** generates a DFF potential. Pacing analysis circuitry **34** extracts portions of the cardiac EGMs of the FF potential, the DFF potential, and/or the NF potential from a time period (e.g., 220 millisecond, etc.) after each pacing test stimulation. The portions extracted from the cardiac EGMs of the FF potential, the DFF potential, and/or the NF potential may be collectively referred to as the “evoked response” of that test pacing stimulation. For each evoked response, pacing analysis circuitry **34** extracts and/or derives physiologically significant features from the evoked response. Pacing analysis circuitry **34** uses classification model **35** to classify each evoked response based on these physiologically significant features. For example, pacing analysis circuitry **34** may classify each evoked response as “selective capture,” “non-selective capture,” “RV capture,” or “no capture.” In some examples, pacing analysis circuitry **34** may additionally or alternatively classify evoked responses to pacing other than the test pacing. For example, pacing analysis circuitry **34** may randomly or pseudo-randomly sample evoke responses after pacing for later analysis via, for example, external device **24**. In some examples, processing circuitry **70** acts in response to the classification of the test pacing. For example, processing circuitry **70** may adjust stimulation voltage for pacing. In one scenario, processing circuitry **70** may reduce stimulation voltage during each subsequent test pacing until the stimulation voltage no longer provides selective capture. Processing circuitry **70** may set the stimulation voltage to be the last voltage that provided selective capture plus, in some examples, an additional voltage margin.

[0040] Classification model **35** is a model that takes, as input, the physiologically significant features (sometimes referred to as the “classification features”), and outputs the classification. Classification model **35** is generated using supervised machine learning. To train the classification model **35**, training sets and validation sets are generated (collectively referred to as “ML sets”). The

ML set are generated by collecting sample evoked responses (e.g., the NF, FF, and DFF potentials of evoked responses) for each classification. Features are extracted or derived from the sample evoked response. These features include quantifiable values, such as the absolute or relative voltages of, for example, the Q-wave, the R-wave, and the S-wave, etc. and the associated timings of these characteristics. Through training, classification model **35** determines the which of the features are physiologically significant features. That is, the physiologically significant features are the features that, through machine learning, the classification model **35** determines are relevant to the classification process. For example, there may be fifteen features that may be extracted and/or derived from the sample evoked responses, but through machine learning, only thirteen of the features may be used to classify evoked responses (e.g., the physiologically significant features). That is, in some examples, the physiologically significant features include some, but not all, of the features of the evoked responses (e.g., the physiologically significant features are a subset of the features of the evoked response, etc.). In such an example, when implemented by IMD **16**, pacing analysis circuitry **34** may only need to extract and/or derive the thirteen physiologically significant features instead of all of the possible extractable and/or derivable features. Different machine learning algorithms may be used, such as a classification algorithm (e.g., a decision tree algorithm, a k-nearest neighbor (KNN) algorithm, etc.) or a regression algorithm (e.g., a linear regression algorithm, a polynomial regression algorithm, etc.). In some examples, a classification machine learning algorithm may produce a decision tree model to classify the pacing based on the features of the evoked responses.

[0041] FIG. **4** depicts an example evoked **400** response illustrating physiologically meaningful features that may be extracted from evoked response **400** that may be used to classify evoked response **400**. In the illustrated examples, evoked response **400** includes a far field (FF) potential **402**, a near field (NF) potential **404**, and a differential far field (DFF) potential **406**. FF potential **402** may be measured between electrode **25A** of lead **22** (sometime referred to as “RVtip”) and housing electrode **4** (sometimes referred to as “Can”). In some examples that include other electrodes, such as an ICD that includes defibrillation electrodes, FF potential **402** may be measured between electrode **25A** and such other electrodes. NF potential **404** may be measured between electrode **25A** of lead **22** and electrode **25B** of lead **22** (sometime referred to as “RVring”). DFF potential **406** may be a first order differential of the FF potential **402**. The physiologically meaningful features may be extracted over window **408** (e.g., **220** milliseconds (ms), etc.) after the pacing is delivered.

[0042] FF potential **402**, FF potential **404**, and DFF potential **406** may have physiologically meaningful features that may be used to classify evoked response **400**. The following are a non-exhaustive list of potential physiologically meaningful features that may be used by classification model **35**. One physiologically meaningful feature may be a time between the pacing and the response of FF potential **402** when the negative deflection is below a certain threshold (sometimes referred to as “T1”, and the zero line is adjusted based on the EGM value before the start of ventricular pacing). One physiologically meaningful feature may be a width, as a measure of time, of the FF potential **402** at negative deflection (e.g., starting and ending when the potential crosses of a pre-defined threshold) (sometimes referred to as “T2”). One physiologically meaningful feature may be a time between the start of ventricular pacing to the positive peak of DFF potential **406** (sometimes referred to as “T4”). One physiologically meaningful feature may be a time between the start of ventricular pacing to the positive peak of NF potential **404** (sometimes referred to as “T6”). One physiologically meaningful feature may be a time between the start of ventricular pacing to the positive peak of FF potential **402** (sometimes referred to as “T.sub.MAX”). One physiologically meaningful feature may be a time between the start of ventricular pacing to the negative peak of FF potential **402** *sometimes referred to as “T.sub.MIN”). One physiologically meaningful feature may be a negative peak amplitude of FF potential **402** from zero line within window **408** (sometimes referred to as “A1”). One physiologically meaningful feature may be an

amplitude of FF potential **402** from negative peak to positive peak within window **408** (sometimes referred to as “A2”). One physiologically meaningful feature may be a positive peak amplitude of DFF potential **406** within a window **408** (sometimes referred to as “A3”). One physiologically meaningful feature may be an absolute peak amplitude from zero line of NF potential **404** (sometimes referred to as “A.sub.MAX”). One physiologically meaningful feature may be a negative slope following the maximum positive peak of FF potential **402** (sometimes referred to as “SP1”). Other physiologically meaningful features may be derived, such as a ratio of T4 to A3, etc. [0043] FIGS. 5A, 5B, 5C, and 5D depict example classifications for evoked responses **500A**, **500B**, **500C**, and **500D** that were classified in accordance with the teachings of this disclosure. Pacing, as evidenced by the evoked responses, may be categorized as ‘selective capture,’ “non-selective capture,” “RV capture,” and “no capture” (sometimes referred to a “no ventricular capture”). In some examples, pacing categorized as “RV capture” may further be categorized as “fusion capture” (e.g., multiple electrical impulses act upon the right ventricular chamber) and “right bundle capture” (e.g., capture of just the bundle of Purkinje fibers **30** of right ventricle **28**). FIG. 5A illustrates an evoked response **502A** indicative of selective capture. FIG. 5B illustrated an evoked response **502B** indicative of non-selective capture. FIG. 5C illustrated an evoked response **502C** indicative of non-selective capture. FIG. 5D illustrated an evoked response **502D** indicative of RV capture.

[0044] FIG. 6 is a conceptual diagram illustrating an example system **600** to generate the classification model of FIG. 1 using supervised machine learning, in accordance with the teachings of this disclosure. Supervised machine learning trainer **602** generates a model (e.g., model **35**) that may operate on IMD **16** to classify evoked responses (e.g., evoked responses **500A**, **500B**, **500C**, and **500D** of FIG. 5). Through supervised machine learning, the machine learning trainer **602** identifies which of the physiologically meaningful features are to be extracted from evoked responses to classify the pacing. Machine learning trainer **602** infers relationships between inputs (e.g., the physiologically meaningful features) and outputs (e.g., the classifications) from a set of labeled pairs. The labeled pairs each include (a) a set of physiologically meaningful features extracted from an evoked response and (b) the classification of that evoked response. The labeled pairs are split into one or more training sets **604** and one or more validation sets **606**. Machine learning trainer **602** infers relationships between the physiologically meaningful features and classifications based on the labeled pairs in training set. In some examples, machine learning trainer **602** uses a classification tree algorithm to produce a candidate model. The candidate model is then evaluated by model evaluator **608** using validations sets **606**. Model evaluator **608** scores the candidate model based on, for example, a percentage of classifications that were correct (sometimes referred to as “true positives”) and a percentage of classifications that were incorrect (sometimes referred to as “false negatives”). Each classification may be scored separately. For example, model evaluator **608** may separately score the “Selective Capture” classification, the “Non-Selective Capture” classification, the “RV Capture” classification, and the “No Capture” classification. A model may be accepted if all classifications satisfy (e.g., are greater than or equal to) a true positive percentage and/or all classifications satisfy (e.g., are less than) a false negative percentage. For example, a candidate model may be accepted if the true positive percentage for all classifications are at least 85 percent and no one classification is falsely identified as another classification more than 5 percent of the time. If a candidate model fails, machine learning trainer **602** may generate a new candidate model with one or more different sets of training data (e.g., labeled pairs may be redistributed randomly between training sets **604** and validation sets **606**, etc.) after tuning the parameters of the machine learning algorithm.

[0045] FIG. 7 is a flowchart of an example method to classify evoked responses, in accordance with the teachings of this disclosure. From time to time, IMD **16** performs a pacing test. For example, IMD **16** may perform a pacing test daily or weekly at a time when patient **14** is likely asleep. IMD **16** uses the pacing test to, for example, determine whether the current stimulation

level of pacing is providing effective capture and whether a different amplitude of stimulation (e.g., an amplitude that uses a lower voltage, etc.) may still provide effective capture. Initially, IMD **16** captures a near field (NF) electrogram and a far field (FF) electrogram for a period after pacing (e.g., 100 ms, 160 ms, 220 ms, etc.) (**702**). IMD **16** generates a differential far field (DIFF) electrogram that is a first order differential of the far field electrogram (**704**). In some examples, the pacing test may provide a series of pacing such that IMD **16** may extract FF, NF, and DFF potentials for multiple beats. In some such examples, one or more of the final pacing stimulus signals may change one or more pacing parameters (e.g., amplitude, pulse width, etc.) such that a new pacing parameter may be compared to existing pacing parameters. IMD **16** extracts the beat(s) (**706**).

[0046] IMD **116** selects the first or next beat (**708**). IMD **16** determines whether an FF electrogram baseline is greater than a threshold (e.g., 0.8 mV, etc.) before the pacing stimulus (**710**). When the FF electrogram baseline is not greater than a threshold (“NO” at **710**) IMD **16** determines that the capture “beat” is an artifact and provides no classification (**712**). Otherwise, when IMD **16** determines whether a FF electrogram baseline is greater than a threshold (“YES” at **710**), IMD **16** extracts features from the FF, NF and DFF electrograms and, in some examples, derive parameters based on features and parameters used by model **35** to classify the beats (e.g., a specific list of features provided when model **35** is downloaded into IMD **16**, etc.) (**714**). IMD **16** classifies the beat using model **35** and the features and parameters (**716**). When there is another beat to classify (“YES” at **718**), IMD **16** selects the next beat (**708**). When there is not another beat to classify (“NO” at **718**), the method ends. In some examples IMD **16** may further take actions in response to the classification, such as adjusting the amplitude and/or pulse width of the stimulation. In some examples, IMD **116** may perform pacing tests with reduced stimulus amplitudes until the beats change from a “selective capture” classification to a “non-selective capture” classification. In some such examples, IMD **116** may set the amplitude of the stimulus based on the lowest amplitude (e.g., with an added safety margin) that was classified as “selective capture.”

[0047] FIG. **8** is a conceptual diagram of a leadless intracardiac pacemaker **800** positioned within the right atrium (RA) of heart **12** for providing ventricular pacing via the His bundle. Pacemaker **800** may include a distal tip electrode **802** extending away from a distal end **812** of the pacemaker housing **805**. Intracardiac pacemaker **800** is shown implanted in the RA of the patient's heart **12** to place distal tip electrode **802** for delivering pacing pulses to the His bundle. For example, the distal tip electrode **802** may be inserted into the inferior end of the interatrial septum, beneath the AV node and near the tricuspid valve annulus to position tip electrode **802** in, along or proximate to the His bundle. Distal tip electrode **102** may be a helical electrode providing fixation to anchor the pacemaker **800** at the implant position. In other examples, pacemaker **800** may include a fixation member that includes one or more tines, hooks, barbs, helices, or other fixation member(s) that anchor the distal end of the pacemaker **800** at the implant site.

[0048] A portion of the distal tip electrode **802** may be electrically insulated such that only the most distal end of tip electrode **802**, furthest from housing distal end **812**, is exposed to provide targeted pacing at a tissue site that includes a portion of the His bundle. One or more housing-based electrodes **804** and **806** may be carried on the surface of the housing of pacemaker **100**. Electrodes **804** and **806** are shown as ring electrodes circumscribing the longitudinal sidewall of pacemaker housing **805** extending from distal end **812** to proximal end **810**. In other examples, a return anode electrode used in sensing and pacing may be positioned on housing proximal end **810**. Pacing of the ventricles, e.g., via the His-Purkinje system, may be achieved using the distal tip electrode **802** as the cathode electrode and either of the housing-based electrodes **804** and **806** as the return anode.

[0049] Cardiac electrical signals produced by heart **12** may be sensed by pacemaker **800** using a sensing electrode pair selected from electrodes **802**, **804** and **806**. For example, a ventricular electrical signal for sensing ventricular R-waves may be sensed using distal tip electrode **812** and

distal housing-based electrode **804**. An atrial electrical signal for sensing atrial P-waves may be sensed using electrodes **804** and **806**. The atrial and ventricular electrical signals may be analyzed for sensing atrial and ventricular events. In some examples, pacemaker **800** is a dual chamber pacemaker configured to deliver atrial pacing pulses using a housing based distal electrode **804** and proximal electrode **806** and deliver ventricular pacing pulses via tip electrode **802** and proximal electrode **806**. Examples of dual chamber intracardiac pacemakers which may incorporate the techniques disclosed herein for controlling ventricular sensing parameters are generally disclosed in U.S. Patent Application Publication No. 2019/0083800 (Yang, et al.), incorporated herein by reference in its entirety.

[0050] Processing circuitry of pacemaker **800** includes pacing analysis circuitry (e.g., pacing analysis circuitry **34**, etc.) and a classification model (e.g., classification model **35**, etc.) configured to classifying evoked responses resulting from pacing. From time to time, shortly after delivering the electrical stimulation (e.g., within 90 to 120 milliseconds, etc.), the pacing signal analysis circuitry measures an evoked response of the heart muscle of heart **12**. The evoked response is measured via an EGM sensed via one or more electrodes of pacemaker **800**. The cardiac signal analysis circuitry may measure the FF potential and the NF potential. The cardiac signal analysis circuitry may also determine the DFF potential. The cardiac EGM that includes an evoked response may include the P-waves, the R-waves, and the T-waves. As described above, the cardiac EGM of an evoked response may include physiologically meaningful features. These physiologically meaningful features are extracted by pacing analysis circuitry. Pacing analysis circuitry then uses classification model to classify the evoked response. In some examples, pacemaker **800** may take actions based on the classification of the evoked response, such as adjusting pacing parameters and/or providing an alert (e.g., to external device **24**)

[0051] The following examples are described herein.

[0052] Example 1A. A system comprises stimulation circuitry configured to generate a pacing stimulus, sensing circuitry configured to sense an evoked response after the pacing stimulus, and processing circuitry configured to: determine classification features from the evoked response, apply the classification features to a classification model, the classification model generated by a machine learning algorithm using one or more test sets comprising a plurality of sample evoked responses for each of a plurality of classifications, and based on the output of the model, classify the evoke response as one of the plurality of classifications.

[0053] Example 1B. The system of Example 1A, wherein the classification features are a subset of possible features of the evoked response.

[0054] Example 1C. The system of 1B, wherein the classification features are determined by the machine learning algorithm from the possible features of the evoked response.

[0055] Example 1D. The system of any of Examples 1A or 1B, wherein the evoked response is indicative of capture of heart muscle in response to the pacing stimulus, and wherein the plurality of classifications includes selective capture, non-selective capture, right ventricular capture, and no ventricular capture

[0056] Example 1E. The system of any of Examples 1A, 1B, or 1D, wherein the classification model is a decision tree model.

[0057] Example 1F. The system of any of Examples 1A, 1B, 1D, or 1E, wherein the processing circuitry is configured to, in response to classifying the evoked response as one of the plurality of classifications, generate an alert.

[0058] Example 1G. The system of any of Examples 1A, 1B, 1D, 1E, or 1F, wherein the processing circuitry is configured to, in response to classifying the evoked response as one of the plurality of classifications, change a voltage level of pacing stimulus.

[0059] Example 1H. The system of any of Examples 1A, 1B, 1D, 1E, 1F, or 1G, including an implantable medical device comprising the stimulation circuitry, the sensing circuitry, and the processing circuitry.

[0060] Example 1J. The system of any of Examples 1A, 1B, 1D, 1E, 1F, 1G, or 1H, wherein the sensing circuitry is configured to sense a near field electrogram (EGM) of the evoked response and a far-field EGM of the evoked response, and wherein the processing circuitry is configured to determine a differential far field EMG.

[0061] Example 1J. The system of Examples 1I, wherein the classification features comprise two or more of (a) a time (T1) between the pacing stimulus and a response of the FF potential when a negative deflection is below a predefined threshold, (b) a width (T2) of the FF potential at the negative deflection, (c) a time (T4) between a start of the pacing stimulus to a positive peak of the DFF potential, (d) a time (T6) between the start of the pacing stimulus to a positive peak of the NF potential, (e) a time (TMAX) between the start of the pacing stimulus to a positive peak of the FF potential, (f) a time (TMIN) between the start of ventricular pacing to a negative peak of the FF potential, (g) a negative peak amplitude (A1) of the FF potential from a zero line within a predefined window, (h) an amplitude (A2) of the FF potential from the negative peak to a positive peak within the predefined window **408**, (i) a positive peak amplitude (A3) of the DFF potential within the predefined window, (j) an absolute peak amplitude (AMAX) from a zero line of the NF potential, and (k) a negative slope (SP1) following a maximum positive peak of the FF potential.

[0062] Example 2A. A method comprising generating, by stimulation circuitry, a pacing stimulus, sensing, by sensing circuitry, an evoked response after the pacing stimulus, determining, by processing circuitry, classification features from the evoked response, applying, by the processing circuitry, the classification features to a classification model, the classification model generated by a machine learning algorithm using one or more test sets comprising a plurality of sample evoked responses for each of a plurality of classifications, and based on the output of the model, classifying, by the processing circuitry, the evoke response as one of the plurality of classifications.

[0063] Example 2B. The method of Example 2A, wherein the classification features are a subset of possible features of the evoked response.

[0064] Example 2C. The method of Example 2B, wherein the classification features are determined by the machine learning algorithm from the possible features of the evoked response.

[0065] Example 2D. The method of any of Examples 2A or 2B, wherein the evoked response is indicative of capture of heart muscle in response to the pacing stimulus, and wherein the plurality of classifications includes selective capture, non-selective capture, right ventricular capture, and no ventricular capture.

[0066] Example 2E. The method of any of Examples 2A, 2B or 2D, wherein the classification model is a decision tree model.

[0067] Example 2F. The method of any of Examples 2A, 2B, 2D, or 2E, comprising, in response to classifying the evoked response as one of the plurality of classifications, generating an alert.

[0068] Example 2G. The method of any of Examples 2A, 2B, 2D, 2E, or 2F, comprising, in response to classifying the evoked response as one of the plurality of classifications, changing a voltage level of pacing stimulus.

[0069] Example 2H. The method of any of Examples 2A, 2B, 2D, 2E, 2F, and 2G, wherein the stimulation circuitry, the sensing circuitry, and the processing circuitry are within an implantable medical device.

[0070] Example 2I. The method of any of Examples 2A, 2B, 2D, 2E, 2F, 2G, and 2H, wherein the evoked response includes a far field (FF) potential, a near field (NF) potential and a differential far field (DFF) potential, and wherein the classification features comprise two or more of (a) a time (T1) between the pacing stimulus and a response of the FF potential when a negative deflection is below a predefined threshold, (b) a width (T2) of the FF potential at the negative deflection, (c) a time (T4) between a start of the pacing stimulus to a positive peak of the DFF potential, (d) a time (T6) between the start of the pacing stimulus to a positive peak of the NF potential, (e) a time (TMAX) between the start of the pacing stimulus to a positive peak of the FF potential, (f) a time (TMIN) between the start of ventricular pacing to a negative peak of the FF potential, (g) a

negative peak amplitude (A1) of the FF potential from a zero line within a predefined window, (h) an amplitude (A2) of the FF potential from the negative peak to a positive peak within the predefined window **408**, (i) a positive peak amplitude (A3) of the DFF potential within the predefined window, (j) an absolute peak amplitude (AMAX) from a zero line of the NF potential, and (k) a negative slope (SP1) following a maximum positive peak of the FF potential.

[0071] Example 2J. The method of any of Examples 2A, 2B, 2D, 2E, 2F, 2G, or 2H, wherein sensing the evoke response includes sensing a near field electrogram (EGM) of the evoked response and a far-field EGM of the evoked response, the method comprising determining, by the processing circuitry, a differential far field EMG based on the far field EMG.

[0072] Example 3A. A computer readable medium comprising instructions, that when executed, cause an implantable medical device (IMD) to: generate, by stimulation circuitry of the IMD, a pacing stimulus; sense, by sensing circuitry of the IMD, an evoked response after the pacing stimulus; determine, by processing circuitry of the IMD, classification features from the evoked response; apply, by the processing circuitry, the classification features to a classification model, the classification model generated by a machine learning algorithm using one or more test sets comprising a plurality of sample evoked responses for each of a plurality of classifications; and based on the output of the model, classify, by the processing circuitry, the evoke response as one of the plurality of classifications.

[0073] The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware or any combination thereof. For example, various aspects of the described techniques may be implemented within one or more processors or processing circuitry, including one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit including hardware may also perform one or more of the techniques of this disclosure.

[0074] Such hardware, software, and firmware may be implemented within the same device or within separate devices to support the various operations and functions described in this disclosure. In addition, any of the described units, circuits or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as circuits or units is intended to highlight different functional aspects and does not necessarily imply that such circuits or units must be realized by separate hardware or software components. Rather, functionality associated with one or more circuits or units may be performed by separate hardware or software components or integrated within common or separate hardware or software components.

[0075] The techniques described in this disclosure may also be embodied or encoded in a computer-readable medium, such as a computer-readable storage medium, containing instructions that may be described as non-transitory media. Instructions embedded or encoded in a computer-readable storage medium may cause a programmable processor, or other processor, to perform the method, e.g., when the instructions are executed. Computer readable storage media may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory (EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a hard disk, a CD-ROM, a floppy disk, a cassette, magnetic media, optical media, or other computer readable media.

Claims

1. A system comprising: stimulation circuitry configured to generate a pacing stimulus; sensing circuitry configured to: sense a near field (NF) electrogram (EGM) of an evoked response after the

- pacing stimulus; and sense a far-field (FF) EGM of the evoked response; and processing circuitry configured to: determine classification features from at least one of the NF EGM and the FF EGM; apply the classification features to a classification model, the classification model generated by a machine learning algorithm using one or more test sets comprising a plurality of sample evoked responses for each of a plurality of classifications; based on output of the model, classify the evoked response as one of the plurality of classifications; and in response to classifying the evoked response as one of the plurality of classifications, change an amplitude level of pacing stimulus.
2. The system of claim 1, wherein the processing circuitry is further configured to: determine a differential far field (DFF) EGM of the evoked response based on the FF EGM; and determine classification features from the DFF EGM; and apply the classification features from at least one of the NF EGM and the FF EGM and the classification features from the DFF EGM to the classification model.
 3. The system of claim 2, wherein the processing circuitry is further configured to: determine classification features from the NF EGM and the FF EGM; and apply the classification features from the NF EGM, from the FF EGM and from the DFF EGM to the classification model.
 4. The system of claim 1, wherein the plurality of classifications includes a selective capture, non-selective capture, and right ventricular capture.
 5. The system of claim 1, wherein the classification model is a decision tree model.
 6. The system of claim 1, wherein the processing circuitry is further configured to, in response to classifying the evoked response as one of the plurality of classifications, generate an alert.
 7. The system of claim 1, wherein the processing circuitry is configured to, in response classifying the evoked response as one of the plurality of classifications, change the amplitude level by changing a voltage level of the pacing stimulus.
 8. The system of claim 1, including an implantable medical device comprising the stimulation circuitry, the sensing circuitry, and the processing circuitry.
 9. The system of claim 1, wherein the evoked response is indicative of capture of heart muscle in response to the pacing stimulus.
 10. The system of claim 2, wherein the classification features comprise two or more of (a) a time (T1) between the pacing stimulus and a response of the FF potential when a negative deflection is below a predefined threshold, (b) a width (T2) of the FF potential at the negative deflection, (c) a time (T4) between a start of the pacing stimulus to a positive peak of the DFF potential, (d) a time (T6) between the start of the pacing stimulus to a positive peak of the NF potential, (e) a time (TMAX) between the start of the pacing stimulus to a positive peak of the FF potential, (f) a time (TMIN) between the start of ventricular pacing to a negative peak of the FF potential, (g) a negative peak amplitude (A1) of the FF potential from a zero line within a predefined window, (h) an amplitude (A2) of the FF potential from the negative peak to a positive peak within the predefined window, (i) a positive peak amplitude (A3) of the DFF potential within the predefined window, (j) an absolute peak amplitude (AMAX) from a zero line of the NF potential, and (k) a negative slope (SP1) following a maximum positive peak of the FF potential.
 11. A method comprising: generating, by stimulation circuitry, a pacing stimulus; sensing, by sensing circuitry, a near field (NF) electrogram (EGM) of an evoked response after the pacing stimulus; sensing, by the sensing circuitry, a far-field (FF) EGM of the evoked response; determining, by processing circuitry, classification features from at least one of the NF EGM and the FF EGM; applying, by the processing circuitry, the classification features to a classification model, the classification model generated by a machine learning algorithm using one or more test sets comprising a plurality of sample evoked responses for each of a plurality of classifications;; based on output of the model, classifying, by the processing circuitry, the evoked response as one of the plurality of classifications; and in response to classifying the evoked response as one of the plurality of classifications, changing an amplitude level of pacing stimulus.
 12. The method of claim 11 further comprising: determining, by the processing circuitry, a

differential far field (DFF) EGM of the evoked response based on the FF EGM; and determining, by the processing circuitry, classification features from the DFF EGM; and applying, by the processing circuitry, the classification features from at least one of the NF EGM and the FF EGM and the classification features from the DFF EGM to the classification model.

13. The method of claim 12, further comprising: determining, by the processing circuitry, classification features from the NF EGM and the FF EGM; and applying, by the processing circuitry, the classification features from the NF EGM, from the FF EGM and from the DFF EGM to the classification model.

14. The method of claim 11, wherein the plurality of classifications includes a selective capture, non-selective capture, and right ventricular capture.

15. The method of claim 11, wherein the classification model is a decision tree model.

16. The method of claim 11, comprising, in response classifying the evoked response as one of the plurality of classifications, changing the amplitude level by changing a voltage level of the pacing stimulus.

17. The method of claim 11, wherein the stimulation circuitry, the sensing circuitry, and the processing circuitry are within an implantable medical device.

18. The method of claim 12, wherein the classification features comprise two or more of (a) a time (T1) between the pacing stimulus and a response of the FF potential when a negative deflection is below a predefined threshold, (b) a width (T2) of the FF potential at the negative deflection, (c) a time (T4) between a start of the pacing stimulus to a positive peak of the DFF potential, (d) a time (T6) between the start of the pacing stimulus to a positive peak of the NF potential, (e) a time (TMAX) between the start of the pacing stimulus to a positive peak of the FF potential, (f) a time (TMIN) between the start of ventricular pacing to a negative peak of the FF potential, (g) a negative peak amplitude (A1) of the FF potential from a zero line within a predefined window, (h) an amplitude (A2) of the FF potential from the negative peak to a positive peak within the predefined window, (i) a positive peak amplitude (A3) of the DFF potential within the predefined window, (j) an absolute peak amplitude (AMAX) from a zero line of the NF potential, and (k) a negative slope (SP1) following a maximum positive peak of the FF potential.

19. The method of claim 11, wherein the evoked response is indicative of capture of heart muscle in response to the pacing stimulus.

20. A non-transitory computer readable medium comprising instructions, that when executed, cause an implantable medical device (IMD) to: generate, by stimulation circuitry of the IMD, a pacing stimulus; sense, by sensing circuitry of the IMD, a near field (NF) electrogram (EGM) of an evoked response after the pacing stimulus; sense, by the sensing circuitry, a far-field (FF) EGM of the evoked response determine, by processing circuitry of the IMD, classification features from at least one of the NF EGM and the FF EGM; apply, by the processing circuitry, the classification features to a classification model, the classification model generated by a machine learning algorithm using one or more test sets comprising a plurality of sample evoked responses for each of a plurality of classifications; based on output of the model, classify, by the processing circuitry, the evoked response as one of the plurality of classifications; and in response to classifying the evoked response as one of the plurality of classifications. change an amplitude level of pacing stimulus.
